Assignment 1 (35 marks) – Deadline (28 Oct 2024)

Question 1 (10 marks)

a) List the element of set$\{x \in \mathbb{R}: x^2 + 5x = -6\}$
b) Write set $\{-4, -3, -2, -1, 0, 1, 2\}$ in set builder notation.
c) Find the cardinality $\left\{\{1,4\}, a, b, \{\{3,4\}\}, \{\phi\}\right\}$
d) Draw the Venn diagram illustrating $\left(A \cap \bar{\bar{B}}\right) \cup \left(A \cap \bar{\bar{C}}\right)$
e) There are are 20 people in your neighbourhood own pets. Five people own cats, rabbit and hamster. Three of them own only hamsters, five own only rabbit and another three own cat. How many total pets in your neighbourhood
f) Let A and B be sets. Use properties of set to show $(A \cup B) \cap \left(A' \cap B \quad\right)' = A$

Question 2 (8 marks)

a) Find the quotient and remainder: 44 is divided by 8
b) Find the quotient and remainder: -111 is divided by 11
c) Is $-29 \equiv 5\ (mod 17)$?
d) Is 37≡3(mod 7)?

Question 3 (4 marks)

a) Convert $(1101\ 1001\ 0101\ 1011)_2$ to octal and hexadecimal representations.
b) Convert $(7206)_8$ and $(A0EB)_{16}$ to a binary representation

Question 4 (13 marks)

a) Show 143 is not prime integer
b) Find the prime factorization of 88
c) gcd*(*1000, 625*)* and lcm*(*1000, 625*)* and verify that gcd*(*1000, 625*)* · lcm*(*1000, 625*)* = 1000 · 625.
d) Use the Euclidean algorithm to find gcd(123,277)
e) Find the greatest common divisor of $2^3 3^5 5^7 7^9 11$ and $2^9 3^7 5^5 7^3 13$